

INCIDENT INVESTIGATION PORTFOLIO

Brute Force Detection & Triage – Windows Domain Controller

Incident Details

INCIDENT ID: IR-ALERT-001

DATE: May 7, 2025

ROLE: Tier 1 SOC Analyst

Scenario

A SIEM alert triggered for user "jdoe" on Domain Controller DC01. Multiple failed logins followed by a successful login at 3:47 AM (outside normal business hours).

Tools Used

- Splunk Enterprise (log ingestion and correlation)
- Windows Event Logs (Event IDs 4625, 4624)
- CSV log analysis

Investigation Summary

FINDING 1: 21 failed logins occurred in 2 minutes

From 03:45:12 to 03:47:13, user jdoe had 21 consecutive failed authentication attempts from source IP 192.168.12.45. This pattern indicates automated brute force tool usage.

FINDING 2: Successful login occurred at 03:47:15

Exactly 2 seconds after the last failure, a successful login was recorded with Logon Type 10 (Remote Interactive / RDP). This suggests the attacker found the correct password.

FINDING 3: Abnormal time of activity

Normal user activity for jdoe occurs between 9 AM and 5 PM. The 3:47 AM login is a critical anomaly.

VERDICT: TRUE POSITIVE – Credential brute force succeeded

SPL Queries Used

```
[1] Identify failure count
index=main user=jdoe event_id=4625
```

```
| stats count by src_ip
```

```
[2] Find successful login  
index=main user=jdoe event_id=4624  
| table _time, src_ip
```

```
[3] Correlate failures → success  
index=main user=jdoe  
| eval outcome = if(event_id=4625, "fail", "success")  
| transaction user maxspan=2m  
| where fail_count > 5 AND success_count > 0
```

Containment Recommendations

IMMEDIATE (within 5 minutes):

- Disable user account for jdoe
- Force password reset at next login
- Block source IP 192.168.12.45 at firewall

ESCALATE TO TIER 2:

- Investigate lateral movement from DC01
- Check for other compromised accounts
- Review RDP logon patterns
- Determine if attacker established persistence

Key Metrics

Time to detect: Immediate (alert triggered)

Time to triage: 8 minutes

Confidence level: 9/10

True positive rate: 100%

Skills Demonstrated

- ✓ SIEM alert triage (Splunk)
- ✓ Windows Event Log analysis (4625, 4624)
- ✓ Pattern recognition (brute force vs legitimate)
- ✓ Timelining and correlation
- ✓ Incident documentation
- ✓ Containment strategy
- ✓ Escalation decision-making

Verification Instructions for Hiring Manager

The original log file is available upon request. You can verify my findings by uploading brute_force_alert1.csv to Splunk and running:

```
index=main user=jdoe | stats count by event_id
```

Expected: 21 x Event ID 4625, 1 x Event ID 4624

Contact Information

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